Dongzhe Zhang

dongzhe.zhang@polimi.it | +39 344 576 7917 | Milan, Italy | LinkedIn | Google Scholar

EDUCATION

Politecnico di Milano & Northwestern Polytechnical University **Apr. 2022 – Expected Apr. 2027**

- Dual Ph.D. Program (Based in Milan from Sep. 2024)
- Lab: Image and Sound Processing Lab (ISPL) – Advisor: Prof. Alberto Bernardini
- Lab: Joint Laboratory of Environmental Sound Sensing – Advisor: Prof. Jianfeng Chen

Northwestern Polytechnical University **Sept. 2019 – Apr. 2022**

- Master of Engineering (Major in Electronic and Communication Engineering)

Shaoxing University **Sept. 2015 – Jun. 2019**

- Bachelor of Engineering (Major in Electronic Information Engineering)

RESEARCH INTERESTS

- **Research Areas**: Deep Learning for Audio, Microphone Array Processing, Sound Source Localization, Speech Enhancement, Distributed Systems for Acoustic Sensing
- **Engineering & Implementation Focus**: Edge AI Implementation, Embedded System Design for Audio Processing, Low-Power Algorithm Design

EXPERIENCE

Tianhe Defense Technology Co., Ltd. Xi'an, China **Jun. 2019 – Oct. 2019**

Embedded Software Engineer

- Engineered the complete firmware for the control unit of an Unmanned Underwater Vehicle.
- Designed an edge-based algorithm for passive underwater target detection and guidance using a single hydrophone.

LianFeng Acoustic Technologies Co., Ltd. Xi'an, China **Oct. 2020 – Oct. 2024**

Technical Co-founder & Acoustic Engineer

- Co-founded an acoustic technology startup, securing initial funding.
- Leading a 5-person algorithm team to develop and commercialize a portfolio of advanced acoustic sensing products.
- Directed the full R&D lifecycle from algorithm design to embedded system deployment and performance validation.

PROJECTS

Unmanned Underwater Vehicle Control & Guidance System **Jun. 2019 – Oct. 2019**

- Developed the complete control unit firmware on an STM32 platform, enabling precise real-time control of propulsion motors and guidance servos.
- Authored and implemented a novel passive target detection and guidance algorithm, designed for a single-hydrophone sensor and deployed on the edge for autonomous operation.

Distributed Sound Event Localization & Detection (SELD) System **Oct. 2020 – Oct. 2022**

- Designed custom microphone arrays and developed core algorithms for sound source localization and anomalous event detection for wide-area monitoring.
- Architected and deployed a distributed sensing system utilizing multiple microphone arrays, successfully validating its localization and detection performance in both indoor and outdoor environments.

Acoustic Camera **Mar. 2021 – Oct. 2024**

- Designed and simulated a novel spiral microphone array in MATLAB to optimize beamforming performance and noise suppression, subsequently assisting with FPGA implementation of the core algorithms.
- Developed a multi-modal geometric calibration system by fusing video and audio streams and implemented a multi-target detection algorithm to resolve distinct sound sources across different frequency bands.

AI-Powered Unattended Ground Sensor (UGS) **May 2021 – Oct. 2024**

- Engineered an end-to-end signal processing pipeline on an STM32 microcontroller, from data acquisition to transmitting processed results to a central controller.
- Designed and deployed a deep learning model for real-time detection of anomalous sound and vibration signals on the microcontroller, achieving system-wide low-power operation through hardware/software co-optimization.

Geometry-Agnostic Speech Enhancement **Jan. 2024 – Present**

- Developed a novel deep learning framework that decouples speech enhancement from array geometry by using an Eigenbeam transform to project sound fields into a universal, orthogonal feature space.

COMPETITIONS

- **2nd** Place, DCASE 2024 Challenge Task 10 (Acoustic-Based Traffic Monitoring)
- Rank 6, DCASE 2023 Challenge Task 3 (Sound Event Localization and Detection in Real Spatial Sound Scenes)
- Gold Award, China International "Internet+" College Students' Innovation and Entrepreneurship Competition, 2021

PUBLICATIONS

- **Zhang D**, Chen J, Bai J, et al. Sound Event Localization and Classification using Wireless Acoustic Sensor Networks in Outdoor Environments[J]. *IEEE Sensors Journal*, 2025.
- **Zhang D**, Chen J, Bai J, et al. Multiple sound sources localization using sub-band spatial features and attention mechanism[J]. *Circuits, Systems, and Signal Processing*, 2025, 44(4): 2592-2620.
- **Zhang D**, Chen J, Huang S, et al. Synthesis-to-real robust training for enhanced sound event localization and detection using dynamic kernel convolution networks[J]. *Applied Acoustics*, 2025, 228: 110267.
- **Zhang D**, Chen J, Bai J, et al. DOA-based multi-node geometry calibration for wireless acoustic sensor network[C]//*2023 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC)*. IEEE, 2023: 1-5.
- **Zhang D**, Bai J, Chen J. JLESS submission to DCASE 2024 Task10: An acoustic-based traffic monitoring solution[J]. *DCASE 2024 Challenge*, Tech. Rep, 2024.
- **Zhang D**, Bai J, Chen J. JLESS submission to DCASE2023 task3: Conformer with data augmentation for sound event localization and detection in real space.[J]. *DCASE 2023 Challenge*, Tech. Rep, 2023.
- Sun W, **Zhang D**, Bai J, et al. JLESS SUBMISSION TO DCASE2024 TASK3: Conformer with Data Augmentation for Sound Event Localization and Detection with Source Distance Estimation[R]. *DCASE2024 Challenge*, Tech. Rep, 2024.
- Han Y, Jianfeng C, **Dongzhe Z**. Passive homing method with reinforcement learning for a single hydrophone[J]. *ACTA ACUSTICA*, 2025, 50(1): 59-67.
- Li X, Chen J, Bai J, et al. Deep learning-based DOA estimation using CRNN for underwater acoustic arrays[J]. *Frontiers in Marine Science*, 2022, 9: 1027830.
- Niu Q, Shi W, Zhang Q, et al. Underwater Passive Sonar Fusion Detection Based on Sensor Bias Estimation and Classification[J]. *Digital Signal Processing*, 2025: 105596.
- Bai J, Wang M, Liu H, et al. Description on IEEE ICME 2024 grand challenge: Semi-supervised acoustic scene classification under domain shift[J]. *arXiv preprint* :2402.02694, 2024.

Submissions Under Review

- **Zhang D**, Chen J, Wang M, Mezza A I, et al. On the Design of Efficient Neural Methods for Geometry-Agnostic Multichannel Speech Enhancement[C]. Submitted to *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
- **Zhang D**, Chen J, Wang M, et al. Co-designing Eigenbeam and Multi-order Encoder for Geometry-Agnostic Speech Enhancement[J]. Submitted to *IEEE Signal Processing Letters*.
- Greco G, **Zhang D**, Pezzoli M, et al. Distributed Acoustic Mixture-of-Experts for Spatial Sound Estimation[J]. Submitted to *IEEE Open Journal of Signal Processing (ICASSP 2026 Track)*.

PATENTS

- Air sonar array device for acousto-optic linkage monitoring and positioning (CN218003722U, Active)
- Wireless low-power consumption ground defense monitoring devices (CN218511891U, Active)
- Pipeline gas leakage amount detection method (CN117515432A, Pending)
- A passive tracking method and device for an underwater target (CN116430370A, Pending)

ADDITIONAL

- Programming Languages: Matlab, Python, C
- AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Librosa, Torchaudio, PyTorch Lightning
- Embedded Systems: Embedded C, STM32(L4/F1/F4/H7), ARM Cortex-M, Keil, STM32CubeIDE, RTOS, I2C, SPI, UART
- Language Skills: English(Professional Working Proficiency, TOEIC-895), Mandarin(Native)